

Chapter 143 — Focal Therapy for Renal Tumours

INTRODUCTION

- **Thermal ablation (TA) use for small renal cell carcinoma (RCC) is rising** in population-based data: rapid growth in the early 2000s, then a plateau at **about one in eight treatments** overall and **almost one-third of treatments** in patients aged 80–89 years.
- **RCC incidence is rising through incidental detection** on cross-sectional imaging performed for abdominal complaints; incidental tumours are small, so **more than 70% of new tumours are clinical stage T1**, with a consequent fall in cancer-specific mortality.
- **Age at diagnosis is rising**, with increasing incidence in patients over 65 years.
- **Rationale for nephron sparing**: chronic kidney disease (CKD) increases all-cause mortality, particularly after nephrectomy, while the cancer mortality of small RCC is low → emphasis on nephron-sparing treatment without oncological compromise.
- **Partial nephrectomy (PN) is the gold standard** for small renal tumours: cancer-specific survival (CSS) >95%, equivalent to radical nephrectomy, with better renal function, superior cardiac outcomes and improved overall survival (OS).
- **PN remains underused**: in Surveillance, Epidemiology, and End Results (SEER) data for 2019, more patients with localised kidney cancer still received radical nephrectomy than PN; PN is used less in the elderly because of concern about complications.
- **2021 American Urological Association (AUA) guideline**: prioritise PN for clinical T1a (cT1a) masses; **consider TA for cT1a masses ≤3 cm**, particularly with substantial comorbidity.
- **Ablative options**: percutaneous cryoablation (CA), radiofrequency ablation (RFA) and microwave ablation (MWA) — extend nephron sparing to highly comorbid patients while minimising morbidity.
- **Approaches**: open, laparoscopic or percutaneous; **modern series are predominantly percutaneous**.
- **Advantages over extirpation**: no tissue dissection or renorrhaphy → **faster convalescence, fewer complications**; post-ablation **renal function similar to or better than after PN**.
- **Indications**: small renal tumours in **high-surgical-risk patients**; **solitary kidney**; **bilateral tumours**; **hereditary syndromes such as von Hippel-Lindau (VHL) disease**, in which repeated treatments and new tumours are expected.

Concept — *Ablation exists because small RCC seldom kills while nephron loss and comorbidity do; treatment is chosen to minimise competing-risk harm, not to maximise oncological radicality.*

CRYOABLATION

Background

- **Cryotherapy**: tissue destruction by extreme cold generated from liquefied gases that cool as they expand (e.g. argon).
- **Historical sequence**: mid-1800s, **Arnott** — ice and salt (–18 to –24 °C) for breast, cervical and skin cancers → liquefied air (–190 °C), solid carbon dioxide (–78.5 °C), liquid oxygen (–182.9 °C), then **liquid nitrogen (–196 °C)** for skin conditions → **1963, Cooper and Lee: first modern cryoprobe** — pressurised liquid nitrogen in a three-channel probe (one inflow, two outflow) at –196 °C, opening cryotherapy to deep, less accessible sites.

- **Argon-based probes — the key breakthrough:** the **Joule–Thomson principle** (rapid expansion of a high-pressure inert gas produces cooling) yields -185.7°C in target tissue; argon systems reach target temperature faster, more efficiently and with a steeper internal thermal gradient.
- **Most commercial CA units are now argon based** (Galil Medical, Endocare and Oncura systems).

Cryotherapy Mechanism of Action

- **Probe design:** an inner chamber delivers compressed argon to a porous membrane at the distal tip, where expansion produces rapid cooling in a closed system; temperatures reach about -160°C during argon evaporation.
- **Tissue death occurs during both freezing and thawing**, by four mechanisms: direct cell injury, vascular injury with ischaemia, apoptosis and immunomodulation.
- **Freeze phase** (closest to probe): extracellular water freezes first $\rightarrow$ osmotic gradient $\rightarrow$ **cellular dehydration and membrane distortion** $\rightarrow$ intracellular ice $\rightarrow$ organelle and membrane damage.
- **Thaw phase:** extracellular osmolarity falls as ice melts $\rightarrow$ water re-enters cells $\rightarrow$ **cellular oedema and lysis**.
- **Vascular injury:** endothelial damage $\rightarrow$ platelet activation $\rightarrow$ thrombosis $\rightarrow$ **tissue ischaemia**.
- **Zonal effect:** **direct injury and necrosis centrally; sublethal temperatures peripherally activate apoptosis**.
- **Immunomodulation:** necrotic cells release organelles, antigens and damage-associated molecular patterns (DAMPs — DNA, heat shock proteins) $\rightarrow$ dendritic-cell antigen presentation $\rightarrow$ T-cell activation $\rightarrow$ further immune-mediated tumour killing.
- **End result:** coagulative necrosis, apoptosis, immunomodulation and eventual **fibrosis and scar**.

Treatment Temperature

- **Lethal threshold is cell- and tissue-specific:** normal renal parenchyma dies at about -19.4°C ; cancer may require as low as -50°C (more fibrous) in small-animal models $\rightarrow$ **preferred target temperature $\leq -40^{\circ}\text{C}$** .
- **Iceball temperature is not uniform:** 0°C at the iceball edge; temperatures consistently below -20°C are reached only 3 mm inside the edge $\rightarrow$ **iceball must extend at least 5–10 mm beyond the tumour margin**.
- **Multiple cryoprobes** may be required, depending on lesion size and probe type and size.
- **Heat-sink phenomenon:** large adjacent vessels dissipate ice formation $\rightarrow$ colder temperatures or longer freezing required.

Freeze-Thaw Cycles

- **Multiple freeze-thaw cycles** enlarge the necrotic zone and improve cure in animal tumour models $\rightarrow$ **double freeze-thaw cycle is the current recommendation**.
- **Thawing is itself lethal**, and may be passive or active:

PASSIVE VERSUS ACTIVE THAW

	Passive thaw	Active thaw
Method	Iceball melts without intervention	Helium forced through the probe; helium warms on expansion (Joule–Thomson), unlike argon, which cools
Speed	Slower	Faster, more efficient

	Passive thaw	Active thaw
Efficacy	Conflicting data on whether active thaw is as effective as passive → practice varies between centres	

- **Common hybrid:** passive thaw between cycles, **active thaw at the end**, so post-treatment bleeding can be addressed sooner.

Duration of Treatment

- **Duration needed for complete cell death in humans is unknown.**
- **Porcine data:** 5-, 10- and 15-minute freezes all produced complete necrosis 5 mm from the probe, but **only ≥10 minutes produced necrosis ≥10 mm** beyond the probe → most contemporary series use **8–10-minute freeze cycles** (see Technique for the alternative 10 + 6–8-minute schedule).

KEY POINTS: CRYOABLATION

- **Argon-based systems** reach treatment temperatures **below –40 °C** via the Joule–Thomson principle.
- Renal CA is performed under **real-time imaging**, with the treatment zone **about 5–10 mm beyond the tumour margin**.
- A **double freeze-thaw cycle**, each freeze about **8–10 minutes**, is the standard of care.

Concept — Cryoablation is a thermal-gradient problem: lethality exists only within the –40 °C isotherm, which lies well inside the visible iceball, so every technical rule — margin, multiple probes, double cycle, duration — exists to push that isotherm beyond the tumour edge.

HYPERTHERMIC MECHANISMS OF TISSUE INJURY

- **Heat-based modalities:** RFA, MWA, laser ablation and high-intensity focused ultrasound (HIFU); they differ in how tissue temperature is raised, and **RFA and MWA are the most commonly used**.
- **Direct and indirect injury** occur, as with CA.

TEMPERATURE THRESHOLDS OF HYPERTHERMIC INJURY

Temperature	Effect	Zone
40–45 °C	Thermal conduction heating → apoptosis, ischaemia from vascular injury, immune and inflammatory responses	Peripheral, sublethal (delayed injury)
≥60 °C	Rapid protein denaturation → coagulative necrosis	Central, lethal
>80 °C	Increased membrane permeability	Central
≥100 °C	Water vaporisation → cell rupture → carbonisation	Adjacent to probe

- **Central zone** (closest to probe, hottest): protein denaturation, necrosis, **mitochondrial dysfunction and inhibition of DNA replication**.

Concept — Heat and cold share the same two-zone architecture — a central lethal zone of direct injury and a peripheral sublethal zone of apoptosis, ischaemia and immune response — differing only in the direction of the temperature gradient.

RADIOFREQUENCY ABLATION

Background

Historical sequence: 1891, **d'Arsonval** — radiofrequency waves heat living tissue → 1928, **Cushing and Bovie** — electrocautery knife (alternating current passes through the patient to a dispersive grounding pad on the lower limb) → 1990, two groups independently — **percutaneous radiofrequency electrodes**

(insulated shaft, exposed metal tip; ablation length set by exposed tip length, but poor circumferential damage limited use to lesions ≤ 1.5 cm) → modern generators and electrodes for larger, complex lesions.

Mechanism of Action

- **Alternating current at 450–1200 kHz** → ionic vibration → molecular friction → heat.
- **Heat is generated within the tissue by ionic agitation, not supplied by the electrode.**
- Heating depends on **energy deposited** and **surrounding blood flow**, which acts as a heat sink.

Variations in Radiofrequency Ablation Equipment

- **Tissue >100 °C chars** → impedance rises → power transmission falls; generators are built to prevent this.

RFA GENERATOR CONTROL: TEMPERATURE VERSUS IMPEDANCE

	Temperature-controlled	Impedance-based
Feedback	Thermal sensors in the electrode monitor tissue temperature	Generator measures tissue impedance
Response	Power adjusted to prevent overheating	Power reduced to prevent charring
Evidence	No clinical data show superiority of either	

RFA ELECTRODE DESIGNS

Electrode	Design	Characteristic strength
Original single monopolar	Exposed uninsulated tip	Tumours ≤ 2 cm only
12-tine umbrella monopolar (Boston Scientific, 1997)	Insulated; tines act as antennas; current redirected from high- to low-impedance prongs	Most spherical ablation volume
9-tine starburst (RITA, AngioDynamics)	Thermistors in 5 tines; energy modulated to average tine temperature	Best reproducibility
"Cool-tip" (Covidien)	Single 17-gauge electrode, internally cooled with chilled saline; impedance based	Largest ablation zone

- These comparative strengths come from porcine liver; **clinical studies favour multitine electrodes** — more complete necrosis, better accuracy and better outcomes.

DRY VERSUS WET RFA; MONOPOLAR VERSUS BIPOLAR RFA

	Mechanism	Advantage	Disadvantage
Wet RFA	Continuous saline infusion near the probe cools the tip; interstitial hypertonic saline forms a virtual "liquid electrode"	Delays charring and impedance rise → larger zones	Less control of ablation size → overtreatment of normal parenchyma
Bipolar RFA	Current flows between two electrodes in the target; heat generated at both and between them	Higher temperatures , larger necrotic zone	Elliptical (not spherical) zone; less accurate → not widely adopted for spherical renal tumours

Treatment Temperature

- **Determinants of ablation:** power delivered, maximum temperature and duration; with low impedance, an expanding sphere of damage spreads from the probe.
- **Current delivered too rapidly** → desiccation at the probe–tissue interface → impedance rise blocks energy transfer and halts heating → **temperatures kept ≤ 105 °C.**

- **Lethal time-temperature combinations** (in vitro, human prostate cells): **45 °C for 60 min, 55 °C for 5 min, 70 °C for 1 min**; even brief exposure **>70 °C** is lethal to RCC in vitro.
- **Temperature-based generators target 90–105 °C**; ablation is **not successful unless ≥70 °C is reached uniformly**.
- **Impedance-based protocol**: start at **40–80 W** → increase **10 W/min** → maximum **130–200 W**, until impedance reaches **200–500 Ω**.
- **Heat sink**: highly vascular targets or large adjacent vessels disperse heat into cooler blood → spared tumour cells and treatment failure.
- **Two cycles separated by a ≥30-second cool-down** are suggested for temperature-based systems, by extrapolation from animal CA data.

Intraoperative Monitoring

- Electrode placement is visible on ultrasound (US), magnetic resonance imaging (MRI) or computed tomography (CT), but **there is no reliable radiographic visualisation of the RFA treatment zone**.
- Contrast with CA: **CA shows an expanding iceball**; the **RFA zone is inferred** from electrode choice, accurate placement and impedance or temperature readings.
- **Independent thermometry**: non-conducting temperature probes at the tumour periphery or deep margin (also usable with CA).
- **Success is judged by** generator feedback, thermal probes and **absence of contrast enhancement** on axial imaging.

KEY POINTS: RADIOFREQUENCY ABLATION

- Radiofrequency energy produces **ionic friction and agitation**; **>70 °C for ≥1 minute** causes irreversible coagulative necrosis and desiccation.
- Treatment is **temperature-based or impedance-based**.
- Real-time monitoring relies on **impedance and temperature**, not visual or radiographic cues.

Concept — RFA is self-limiting: heat is made in the tissue by ionic friction, so anything that raises impedance (charring, desiccation) or carries heat away (blood flow) stops the ablation — every equipment variation is a device for keeping impedance low and the zone spherical.

MICROWAVE ABLATION

Background

- **Microwaves**: electromagnetic radiation; ablative frequencies **915 MHz or 2.45 GHz**.

Historical sequence: 1930s — tissue heating proposed, generators inadequate → 1930s–1940s — progressively more powerful magnetron tubes → after the Second World War, a Massachusetts Institute of Technology magnetron (400 W at 3000 MHz) → 1946 — Mayo Clinic animal exposure studies → late 1970s — antennas for local tissue destruction.

GENERATIONS OF MICROWAVE SYSTEMS

Generation	Features
Early antennas	Proximal shaft heating → teardrop-shaped zone and risk of fistulous tract
First	No active antenna cooling; low power, short treatments
Second	Antenna cooling; generators still of limited power

Generation	Features
Current	High power plus antenna cooling; gas-cooled antennas form a small freeze zone near the tip that anchors the antenna against migration without compromising the ablation zone

Mechanism of Action

- **Dielectric hysteresis (rotating dipoles):** polar molecules continuously realign with a field alternating billions of times per second → kinetic energy → heat; frequencies **915 and 2450 MHz**.
- **Water is polar** → high-water tissues (most tissues and tumours) heat best.
- **Three components:** generator (**magnetron or solid-state amplifier**), coaxial cables, antenna.
- **Antennas radiate energy — no grounding pads or return electrodes.**
- **Microwaves propagate through desiccated tissue and blood**, so ablation is not limited by charring; tissue often exceeds **100 °C**.

PURPOSE OF PROBE COOLING: RFA VERSUS MWA

RFA electrode cooling	MWA antenna cooling
Prevents tissue charring, which otherwise limits ablation	Protects the insertion track from unwanted ablation and permits higher power

- **Advantages over RFA: faster heating** → **less susceptible to heat sink** (important in the highly vascular kidney); **active heating up to 2 cm** from the antenna versus a few millimetres for RFA (the rest by conduction); **multiple antennas can run simultaneously** for larger tumours; larger zones and higher temperatures than equivalent RFA applicators in porcine kidney.
- **Disadvantages:** energy harder to distribute; **bulkier coaxial cables**, more prone to heating; rapid high temperatures endanger **adjacent sensitive structures** in poorly selected tumours.

Treatment Duration and Power

- **Current devices:** 915 MHz or 2.45 GHz, **45–150 W**, gas- or fluid-cooled antennas; high temperatures allow larger tumours to be treated.
- **Temperature control versus power control:** no significant difference in ablation geometry (porcine kidney).
- **Time and power:** increasing either enlarges the zone, but **power beyond 45 W adds nothing** → porcine optimum **45 W for 10 minutes**; longer ablation time consistently gives larger zones.
- **Multiple simultaneous antennas** exploit overlapping fields to enlarge the zone.
- **Settings are largely manufacturer-derived** → the operator must know the zone shape and size produced by each time–power combination on their own system.
- **No single strategy is evidence-based;** for clinical T1 masses, typically **5–15 minutes at 45–65 W**; authors' standard: **65 W for 5 minutes** (NeuWave system).

KEY POINTS: MICROWAVE ABLATION

- MWA heats tissue by **dielectric hysteresis**.
- Tissue temperatures often **exceed 100 °C** without desiccation or charring limiting the zone.
- MWA is **less susceptible to heat sink** than CA and RFA.
- **Zone size is directly proportional to time** of energy application.

COMPARISON: THE THREE ABLATIVE ENERGIES

	Cryoablation	Radiofrequency ablation	Microwave ablation
Physics	Argon expansion (Joule–Thomson)	Alternating current; ionic friction	Electromagnetic field; dielectric hysteresis
Lethal target	$\leq -40\text{ }^{\circ}\text{C}$	$\geq 70\text{ }^{\circ}\text{C}$ uniformly; generator target 90–105 $^{\circ}\text{C}$	Often $>100\text{ }^{\circ}\text{C}$
Limited by charring	No	Yes	No
Grounding pads	No	Yes (monopolar)	No
Heat-sink susceptibility	Yes	Yes	Least
Zone visible during treatment	Yes — iceball on CT, MRI, US	No — impedance/temperature only	Gas bubbles on US

Concept — MWA's advantages all follow from radiating energy rather than conducting current: no circuit for charring to break, no grounding pad, active heating at a distance, and speed that outruns the heat sink — and its main hazard is that same speed next to adjacent organs.

PERCUTANEOUS ABLATION SURGICAL TECHNIQUE

Patient Selection and the Role of Preablation and Postablation Biopsy

Renal mass management algorithm (Fig. 143.3): initial counselling (evaluate age, comorbidity, CKD risk, tumour size and location; educate on tumour risk versus competing comorbidity; shared decision making) → **surgery**, nephron sparing if feasible (younger, few comorbidities, preference) or **watchful waiting / active surveillance** (significant comorbidity, preference) or **renal mass biopsy** → **second counselling** (review biopsy, staging, options) → surgery, surveillance or **percutaneous ablation** — **offered only after biopsy has been performed and discussed**.

- **2021 AUA guideline: biopsy universally, before (preferred) or at the time of ablation**, to establish pathology and guide surveillance; the authors biopsy all candidates beforehand.
- **Biopsy avoids** unnecessary treatment and post-ablation imaging surveillance of benign masses.

BIOPSY TIMING: SEPARATE PRIOR SESSION VERSUS SAME SESSION AS ABLATION

Separate prior session (preferred)	Same session, immediately before ablation
Fewer non-diagnostic results (biopsy can be repeated)	Non-diagnostic biopsy cannot be repeated before treatment
Higher proportion of ablations are RCC	More benign tumours ablated
Better informed consent	Consent obtained without histology

- **Routine immediate post-ablation biopsy adds nothing** beyond modern CT or MRI; biopsy is reserved for a **new enhancing lesion suspicious of local recurrence** on follow-up.

Collaboration Between Urology and Radiology

- **Successful programmes need both specialties**; in a United States academic survey most centres performed few ablations per month, radiologists alone performed them most often, a combined urologist–radiologist approach was common, and urologists alone least often.
- **Urologist**: pre-procedure counselling on all kidney cancer options and long-term clinical cancer surveillance. **Radiologist**: imaging before, during and after. Both may perform the procedure → a symbiotic collaboration.

Procedure

Approach and anaesthesia

- **Laparoscopic ablation has declined sharply since 2010 while percutaneous ablation has risen** → the vast majority is now percutaneous (less morbidity and cost).
- **Conscious sedation with local anaesthesia or general endotracheal anaesthesia**, outpatient or with 23-hour observation.
- **Outpatient suitability**: tumour of minimal complexity ≤ 3 cm, low bleeding risk (**no anticoagulation, no post-ablation haematoma**), no significant comorbidity.
- **General anaesthesia controls respiration during probe placement** → **better outcomes**.
- **High-frequency jet ventilation**: short high-flow pulses through a small-calibre tracheal catheter, **about 120 breaths/min**; small tidal volumes reduce renal motion; needs a specialised ventilator.
- **Conscious sedation** reserved for highly selected patients who cannot tolerate intubation.

Positioning and guidance

Intravenous prophylactic antibiotics → **prone or modified flank position on the CT gantry** (chosen by tumour location) → probe placement under **US** (or CT or MRI) guidance → **non-contrast CT to confirm probe position**.

Protecting adjacent structures

- **Hydrodisplacement** (Fig. 143.4): spinal needle into the plane between tumour and structure (bowel, ureter, liver, spleen, pancreas) → inject **sterile saline, or 5% dextrose in water (D5W) if using RFA**; contrast may be added for CT fluoroscopic confirmation of separation (e.g. colon displaced >1 cm).
- Saline may be added slowly during ablation to prevent overheating; **avoid excess fluid in poor cardiac function** (absorption).
- **Pyeloperfusion** for central tumours or tumours near the ureter: pre-placed ureteral access catheter irrigated with **cool saline for RFA/MWA, warm saline for CA** — always opposing the ablative temperature — to protect the collecting system.
- **Probe number and duration follow lesion size and manufacturer recommendations**; all modalities require intraprocedural monitoring.

Cryoablation

- **CA's advantage: the iceball is visible** on CT or MRI and on real-time US (ice–tissue interface); on CT it is a **distinct hypodense zone**.
- Probe tips reach **-140 to -190 °C** with a steep gradient to **0 °C at the iceball edge**; death is reliable at -40 °C → **iceball 5–10 mm beyond the tumour margin**.
- **Two freeze-thaw cycles**. No standard duration provided the intended zone is reached; commonly **10 minutes for the first cycle and 6–8 minutes for the second** — animal data show inadequate necrosis at 5 minutes and more tissue fracture at 15 minutes, so 10 minutes is the compromise.
- Each freeze is followed by an **active (helium) or passive thaw** → after the second thaw, the probe is **gently twisted and removed only if there is no resistance**.
- **Multiple probes 1–2 cm apart** create overlapping iceballs and an adequate edge margin; single probes rarely ablate most renal masses completely.
- Tumours **>4 cm** have been ablated, but **haemorrhage increases with tumour size and multiprobe CA**.
- **Contrast-enhanced CT after treatment** assesses completeness and excludes complications.

Radiofrequency ablation

- Efficacy is monitored by **tissue temperature or impedance** — single or multitined electrodes with thermistors, or multiple single-shaft impedance electrodes.
- **Electrode positioned for complete coverage plus a ≥ 5 mm peritumoral margin** (Fig. 143.6).
- **Radiation reduction:** confirmatory scans limited to three or four 3-mm images centred on the lesion at lower tube current (70 versus 150 mA).
- Two cycles at **105 °C**, separated by a **30-second cool-down**; **passive temperature ≥ 70 °C in each quadrant** during cool-down confirms no significant heat sink.
- **Contrast-enhanced helical CT** after treatment; inadequately treated areas → reposition and repeat. **Tract ablation** on withdrawal (manufacturer dependent). Larger lesions: multiple single electrodes in overlapping ablations.

Microwave ablation

- **Antennas placed tangentially under US guidance**, so the tip neither points towards nor penetrates the collecting system; the smaller, more spherical **Precision PR antenna** is preferred to the older LK antenna.
- Antennas are **fixed by cryogenic cooling** → non-contrast CT confirms position → ablation by **time and power** (e.g. 65 W for 5 minutes) → continuous US shows **gas bubbles progressively covering the tumour** (Fig. 143.7) → **contrast CT: no enhancement in the zone including a ≥ 5 mm margin**, and exclusion of haematoma.
- Larger tumours: antenna repositioning. **Poor renal function: unenhanced CT plus contrast-enhanced US.**

INTRAPROCEDURAL PROTOCOLS BY MODALITY

	Cryoablation	Radiofrequency ablation	Microwave ablation (NeuWave Certus 140)
Margin	Iceball 5–10 mm beyond tumour	≥ 5 mm	≥ 5 mm non-enhancing
Probes by size	Multiple, 1–2 cm apart	Tine deployment <2 cm → 5 min; 2–3 cm → 7 min; 3–4 cm → 8 min	≤ 2.5 cm → 1 antenna; 2.5–3.5 cm → 2; 3.5–4.5 cm → 3
Cycles	Two freeze-thaw	Two, 30-s cool-down between	Single time–power application
Endpoint check	Iceball on imaging; post-procedure contrast CT	Quadrant temperature ≥ 70 °C ; contrast CT	Gas bubbles on US; contrast CT

Radiographic Interpretation of Success

- **Complete loss of contrast enhancement on follow-up CT or MRI = treatment success.**
- **2021 AUA: contrast-enhanced cross-sectional imaging within 6 months of ablation.**

INCOMPLETE ABLATION VERSUS LOCAL RECURRENCE

Incomplete ablation	Local tumour recurrence
Persistent enhancement in any part of the lesion on initial imaging	Initial complete loss of enhancement , then later enlargement and/or enhancement
Consider repeat ablation	Biopsy and/or treatment: surveillance, repeat ablation or extirpation (Fig. 143.8)

POST-ABLATION IMAGING APPEARANCE BY MODALITY

Modality	Appearance	Basis
CA	>50% reduction in size within the first year in most lesions	Cellular breakdown and phagocytosis
RFA	Minimal contraction; distinctive fibrotic halo (circular demarcation) after percutaneous treatment (Fig. 143.6)	Foreign-body giant-cell fibrotic response
MWA	Non-enhancing low-attenuation zone on CT; contracts during ablation and continues to contract	Tissue desiccation
MRI (all)	T1 hypointense zone without gadolinium enhancement	—

- **Enlargement of a lesion, regardless of modality or enhancement, is recurrence** until proved otherwise.

Recommended Radiographic Follow-Up Protocol

- **AUA: abdominal CT or MRI with and without intravenous contrast plus chest imaging within 6 months, then at 12, 24, 36, 48 and 60 months;** beyond 5 years, **shared decision making** dictates further imaging.
- **Risk-adjusted follow-up** may be considered, especially for larger or high-grade tumours.
- **US is not routine** after ablation unless a contrast-enhanced US protocol is in place.

KEY POINTS: TUMOUR ABLATION TECHNIQUE AND FOLLOW-UP

- **Biopsy before the ablation session** is safe, increases histological diagnosis and reduces benign tumour ablation.
- Ablation must achieve an **adequate margin**, which depends on the modality.
- **Complete loss of enhancement** on CT or MRI indicates complete destruction and success.
- **Enlargement** (any modality) or **enhancement** in the zone → local recurrence or incomplete ablation.
- **AUA: CT or MRI with and without contrast plus chest imaging at 6, 12, 24, 36, 48 and 60 months;** thereafter shared decision making.

Concept — Ablation leaves the tumour in situ, so the pathological margin of surgery is replaced by an imaging margin: enhancement is the only surrogate of viability, which is why histology must precede treatment and why surveillance imaging is structured and prolonged.

ONCOLOGIC OUTCOMES

Oncologic Outcomes of Thermal Ablation and Limitations of the Current Literature

- Similar metastasis-free survival (MFS) and CSS have prompted claims that PN and TA are equivalent for cT1 RCC — **not strongly supported: no appropriately designed trial has compared TA with surgery.**
- **Existing TA data are biased in both directions:**

BIASES IN THE THERMAL ABLATION LITERATURE

Bias	Nature	Effect on TA results
Non-standard outcome definitions	After PN, local residual/recurrence (≈2%) = tumour on post-operative imaging. After TA, residual tumour may be classed as lack of technical success, incomplete ablation or local tumour progression , depending on imaging timing, which varies widely; surveillance time points are inconsistent	Non-comparable with surgery

Bias	Nature	Effect on TA results
Older, sicker cohorts	Older guidelines restricted TA to non-surgical candidates; TA patients are older with higher comorbidity scores → worse OS despite similar CSS ; many die of other causes before progression is detected	Obscures cancer outcomes
Grade under-reported	Percutaneous biopsy under-samples grade (sampling error, heterogeneity); a meaningful minority of 1–3 cm RCC is high grade in surgical series, far fewer in TA series → almost no data on high-grade tumours	Favours TA
Benign tumours included	Many series lack biopsy-proven RCC	Favours TA
Occult pT3a	Some cT1 tumours have radiologically undetected fat or small-vein invasion; upstaging to pT3a markedly increases recurrence in ≤4 cm surgical series, yet TA is compared with pT1a surgical outcomes	Against TA
Growth-selected treatment	Comorbid patients on active surveillance are treated only on growth (a marker of metastatic potential) → higher-risk tumours treated	Against TA

- **Immediate post-ablation contrast imaging** permits immediate retreatment of residual enhancing tumour and **should be standard practice**; consensus **risk-adjusted follow-up** protocols should be developed.
- **Average age at RCC diagnosis in the United States is 64 years**; TA cohorts are typically about a decade older than PN cohorts.
- **Long natural history — the most important limitation**: SEER kidney cancer mortality ≈5% at 10 years for tumours ≤4 cm; slow growth of many small masses established **active surveillance** as safe for comorbid patients, and a prospective registry found near-identical CSS with upfront PN and initial surveillance.
- **European Organisation for Research and Treatment of Cancer (EORTC) 30904** — PN versus radical nephrectomy for masses ≤5 cm, 534 patients, median follow-up **9.3 years**: only **21 metastatic progressions and 12 RCC deaths** → too few events for definitive conclusions.
- **Implication**: trials comparing small-RCC treatments need **very large cohorts followed for many years**.

Local Recurrence-Free Survival

- **Meta-analyses: local recurrence-free survival (LRFS) is better after PN than TA**, but the **difference disappears once salvage re-ablation is counted**.
- Meta-analysis of TA versus PN for cT1: **higher all-cause mortality with TA**, but **no significant difference in local recurrence or metastasis**; a further meta-analysis found equal 5-year local recurrence.
- **Caveat**: many tumours were not biopsy-proven RCC → inclusion of benign tumours biases towards ablation.
- **Tumour size governs success**: RFA LRFS markedly worse for tumours ≥3 cm than <3 cm; LRFS and disease-free survival (DFS) worse for **T1b (>4 cm)** than T1a; **CA for T1b (>4 to ≤7 cm)** gave substantially more recurrence than PN.
- **MWA for biopsy-proven cT1b RCC**: no difference from PN or radical nephrectomy in MFS, CSS or LRFS (versus PN); **lower OS**, reflecting a more comorbid cohort.
- **Only the National Comprehensive Cancer Network (NCCN) offers TA for cT1b**, in select non-surgical patients; all other guidelines reserve TA for cT1a.

TABLE 143.1 CONSENSUS GUIDELINE RECOMMENDATIONS REGARDING USE OF THERMAL ABLATION FOR LOCALISED RENAL CELL CARCINOMA MANAGEMENT

Society	Size ceiling	Patient group and qualifier
NCCN (2022)	cT1; cT1b in select patients	Option for cT1; cT1b only if not eligible for surgery ; may need multiple treatments to match surgical outcomes
European Association of Urology (EAU) (2022)	TA not routinely >3 cm; CA not routinely >4 cm	Offer active surveillance or TA to frail and/or comorbid patients with small renal masses
AUA (2021)	cT1a ≤3 cm	Consider TA as an alternative; percutaneous preferred over surgical approach whenever feasible
American Society of Clinical Oncology (ASCO) (2017)	Not specified	Percutaneous TA an option where complete ablation is achievable

Data from Motzer et al., 2022; Ljungberg et al., 2022; Campbell et al., 2021; Finelli et al., 2017.

- **What the table tests:** the size ceiling — **3 cm for heat-based TA (AUA, EAU), 4 cm for CA (EAU); NCCN is the only body extending TA to cT1b.**
- **Trap:** TA is an option across guidelines, but **never the default for fit patients** — AUA prioritises PN and frames TA as an alternative.

Metastatic-Free Survival

- **MFS:** freedom from disease outside the treated kidney or renal fossa after primary treatment.
- **Meta-analyses:** no difference in MFS between extirpative surgery and TA.
- Multiple series with **>5 years' follow-up for RFA and CA** suggest **durable outcomes beyond 5 years**; MWA, being newer, has **3–5-year data showing excellent MFS**, comparable with other modalities and surgery.

TABLES 143.2–143.4 OUTCOMES AFTER RFA, CA AND MWA OF BIOPSY-PROVEN RENAL CELL CARCINOMA (SERIES CONSOLIDATED BY MODALITY)

Modality	Approach in series	Follow-up	Tumour size	LRFS	MFS	DFS
RFA (Table 143.2, intermediate to long term)	Percutaneous and laparoscopic	≈2–6.5 years	≈2–3 cm	5-year ≈80–100%, most ≥90%	5-year ≈88–100%	5-year ≈80–100%
CA (Table 143.3, intermediate to long term)	Mostly laparoscopic; one large percutaneous series	≈3–8 years	≈2–2.5 cm	5–10-year ≈80–97%	≈95–100%	5–10-year ≈80–91%
MWA (Table 143.4, short to intermediate term)	Percutaneous only	≈1.5–5.5 years	≈2.3–2.9 cm (one cT1b series)	3–5-year ≈88–98%	≈94–100%	≈83–85%

Data from institutional series, 2008–2022.

- **What the tables are for:** showing that ablation of **biopsy-proven** small RCC is **durable** — LRFS mostly around 90% or better and MFS near 100% — across all three energies.
- **Pattern:** RFA and CA carry the long follow-up; **MWA is entirely percutaneous and has the shortest follow-up**; tumours in all series average 2–3 cm.

- **Trap:** in one RFA series, **no recurrence occurred below 4 cm**; in one CA series, recurrences treated by salvage were excluded from the recurrence analysis, inflating LRFS — series figures are not directly comparable.

Cancer-Specific Survival

- **CSS:** proportion surviving without death from RCC.
- **CA:** CSS 80–100% at up to 10 years; **RFA:** ≈98%.
- **Meta-analyses:** no difference in CSS between extirpative surgery and TA.

Overall Survival

- Ablation patients are older and more comorbid → **OS 75–85% at 5 years, falling to 54–64% at 10 years.**
- Comparative studies show **lower OS after TA than PN**, attributable to comorbidity, not cancer.

Comparison of Ablation Modalities

- **No prospective trial has compared ablation modalities;** observational comparisons are confounded by selection, tumour size and location, technique, approach and experience.
- **Network meta-analysis: no modality oncologically superior; MWA had the lowest complication rate.**
- Single-centre and single-institution comparisons: similar 2-year DFS, MFS and CSS for MWA, RFA and CA; **no significant LRFS difference between RFA and CA** in biopsy-proven RCC; **no CSS difference** in long-term cT1a data (benign masses included).
- **Summary:** no difference in local recurrence, metastatic progression, CSS or OS between modalities; AUA, ASCO, EAU and NCCN accept RFA, CA and MWA, with choice by **institutional preference and tumour characteristics.**

Laparoscopic Versus Percutaneous Renal Tumour Ablation

LAPAROSCOPIC VERSUS PERCUTANEOUS ABLATION

	Laparoscopic	Percutaneous
Technical and oncological efficacy	No significant difference	
Morbidity, procedure time, stay, cost	Higher	Lower morbidity, faster, shorter stay, cheaper
Status	Reserved for prior surgical scarring or failed hydrodisplacement (see Perioperative Outcomes)	Standard of care

KEY POINTS: ONCOLOGIC OUTCOMES

- Interpretation is limited by **non-standard outcome measures, selection bias, biopsy limitations, benign tumour inclusion** and the **indolence** of many small renal masses.
- **Local recurrence is more common with TA than PN**, but comparable once additional ablation is counted.
- **Tumour size** affects ablation success and LRFS.
- **No difference in MFS or cancer mortality** between TA and PN for small renal masses.
- Guidelines accept **RFA, CA and MWA**; choice depends on institutional preference and tumour characteristics.

Concept — Every comparison of ablation with surgery is confounded by who is ablated — older, comorbid patients with under-graded, sometimes benign or occult-pT3a tumours, in a disease that rarely kills within

a decade — so the only robust conclusions are that local recurrence is slightly higher and salvageable, and that cancer-specific outcomes do not differ.

PERIOPERATIVE OUTCOMES

- **TA versus surgery:** shorter stay, less blood loss, lower transfusion rates; in one randomised trial, **MWA had fewer complications than surgery.**
- **Meta-analytic observations:** TA cohorts are older and more comorbid (lower OS, higher other-cause mortality, similar kidney-cancer mortality) and so at higher complication risk; **percutaneous carries fewer complications than laparoscopic** ablation; complications should be graded by **Clavien–Dindo**; major complications are uncommon.
- **Major complications (Clavien–Dindo ≥ 3) in $\leq 5\%$ after CA, RFA or MWA for small tumours; major urological complications fewer than after minimally invasive or open PN,** depending on the population.
- **No direct comparison of urological complications between CA, RFA and MWA.**
- **Most events are minor and self-limiting:** small asymptomatic haematomas and transient urinary retention, often unreported.
- **Laparoscopic ablation has more complications,** a substantial share attributable to the laparoscopic technique itself; complications fall with centre experience.

PREDICTORS OF COMPLICATIONS

Tool or factor	Content	Finding
RENAL nephrometry score (Kutikov and Uzzo, 2009)	Tumour complexity: low 4–6, moderate 7–9, high 10–12	Complications rise with complexity (none at low complexity in laparoscopic CA); high complexity carries substantial major-complication risk
American Society of Anesthesiologists (ASA) score >3	Patient fitness (EuRECA multicentre European laparoscopic CA registry)	About threefold increase in overall complications
(MC)² score	CA-specific; adds prior myocardial infarction or diabetes to tumour factors	Externally validated for CA; not predictive after MWA

Pain and paraesthesia

- **Insertion-site pain or paraesthesia,** historically common, has fallen: **modern cryoprobes are thermally insulated along the shaft,** reducing freezing burns.
- **RFA:** active only at the distal tip; **tract ablation only long enough to clear the kidney and Gerota fascia,** to avoid nerve injury.
- **Grounding-pad burns** (rare): place pads at the **same level on the posterior thighs, perpendicular to the long axis of the thigh** — returning energy takes the shortest arc, so this maximises dissipation area.
- **MWA:** shaft cooling, no grounding pads → pain or paraesthesia rare.
- **Posterior tumours:** injury to nerves on the posterior abdominal wall → **self-limiting neuralgia or neuropraxia;** avoid by positioning so the tumour falls away from the wall, or by hydrodisplacement.
- **Post-procedure pain is minimal** (low patient-reported scores on days 1–2) — a clear recovery advantage over surgery.

SPECIFIC COMPLICATIONS: RISK, PREVENTION AND MANAGEMENT

Complication	Risk factors and presentation	Prevention	Management
Haemorrhage — the most common	Post-procedure haematoma frequent on imaging; transfusion ≈2–3% (falling). Main risk: multiple probes for larger masses; CA theoretically higher risk (no coagulative heating), no comparative data	Allow full thaw before CA probe removal (premature removal → tumour fracture, bleeding); avoid intercostal arteries	Bleeding at needle placement → start ablation (RFA/MWA coagulate); laparoscopic: haemostatic agents + pressure; persistent → selective transarterial embolisation ; abdominal wall haematoma → serial imaging if rapidly expanding, serial haematocrit if stable; prophylactic embolisation for large masses of uncertain benefit
Adjacent organ injury	Tumours <1 cm from an organ , typically anterior (duodenum, colon, pancreas)	Cross-sectional planning, positional imaging, hydrodisplacement , blunt-tip needle	Prior surgical scarring or failed displacement → laparoscopic approach
Collecting system and ureter	Deep thermal extension for central tumours ; haematuria, urinoma ± obstruction, urinary fistula (Fig. 143.9); permanent damage → calyceal or ureteric obstruction (pelviureteric junction or distal); rarely delayed cutaneous fistula	Locate ureter; stent with pyeloperfusion; CA theoretically spares the collecting system → preferred for central tumours ; tangential antenna placement	Haematuria: conservative, significant bleeding → selective angioembolisation; obstruction or leak: conservative or indwelling ureteral stent ; significant urinoma → percutaneous drain
Pleural injury	Probes above the twelfth rib for upper-pole lesions → pneumothorax or haemothorax	Recognise intraprocedurally (dyspnoea, imaging)	Simple pneumothorax → needle aspiration at end of case; larger → intraprocedural blood patch ; chest tube rarely , only if symptomatic; post-operative chest pain or dyspnoea → upright chest radiograph
Bowel injury (exceedingly rare)	Tumour close to bowel	Saline displacement to create a working space	General surgical consultation; controlled colonephric fistula → ureteral stent + low-residue diet ; persistent or colocutaneous fistula → diversion or trial of total parenteral nutrition; perforation with peritonitis → prompt exploration
Infection	<1% without large haematoma or urinoma, but may be lethal; risk: chronic urinary colonisation (e.g. ileal conduit), active infection; presents 1–6 months later as chronic drainage or retroperitoneal abscess	Screen by urine culture and treat urinary tract infection (UTI) beforehand; single-dose prophylaxis (authors' practice — no consensus)	—

KEY POINTS: COMPLICATIONS AFTER TUMOUR ABLATION

- Common complications: **insertion-site pain and paraesthesia, UTI, damage to surrounding structures, self-limiting haematuria.**
- **Post-ablation pain is minimal** with modern devices.

- **Major urological complications are fewer with ablation (≈5%)** than with laparoscopic or open PN.
- **Haemorrhage is the most commonly cited major adverse event.**
- **No significant difference** in urological complications among CA, MWA and RFA (no direct comparative data).
- Complication risk is predicted by **tumour complexity — size and location.**

Concept — *Ablative complications are geographical: they follow the probe path and the thermal field, so prevention is planning — probe count, trajectory, displacement of neighbouring organs and protection of the collecting system — rather than intraoperative rescue.*

NEW ABLATION MODALITIES

High-Intensity Focused Ultrasonography

- **Principle:** part of an acoustic wave's energy is absorbed as heat; focusing with a shaped transducer raises the **focal point above the lethal threshold while sparing adjacent tissue.**
- At $>3500 \text{ W/cm}^2$, **cavitation and microbubble formation** produce very high temperatures and a mechanical "shock wave" effect similar to extracorporeal shock-wave lithotripsy.
- **Unique among TA technologies: entirely non-invasive and extracorporeal** → minimal or no risk of tumour seeding, haemorrhage or urinary extravasation.
- **Technique:** one transducer treats and monitors; the beam is focused on the zone under real-time guidance, then refocused to ablate overlapping volumes and "paint" a larger region.
- **Evidence:** early trials showed feasibility, but **ablate-and-resect studies found incomplete treatment in every specimen**; a small transcutaneous series found viable tumour on early MRI in about half, though most with longer follow-up showed involution.
- **Reasons for failure:** poor targeting from **respiratory movement**; **acoustic interference** (shadowing, reverberation, refraction); lack of effective intraoperative monitoring.
- **Status:** outcomes have stagnated and are **inferior to current ablative technologies** → **investigational.**

Histotripsy

- **Focused ultrasound with a mechanical (non-thermal) mechanism**, under real-time imaging guidance — the key contrast with HIFU.
- Investigated preclinically, especially in liver and kidney models.
- **HOPE4KIDNEY:** single-arm trial of safety and technical success in primary solid renal tumours, accruing from 2023.

Radiation Therapy

- **Conventional external-beam radiotherapy was historically ineffective** for RCC: perceived inherent radioresistance, **limited parenchymal tolerance**, **scatter** damaging surrounding tissue, difficult target localisation.
- **Stereotactic body radiation therapy (SBRT) = stereotactic ablative radiotherapy (SABR):** a high dose delivered in a single or few fractions with high precision.
- **Technique:** three-dimensional coordinates; **real-time tracking and correction for respiratory and organ movement** without interrupting treatment → minimal scatter, focal high dose, ablation **without compromising overall renal function.**
- **Porcine kidney (CyberKnife): 24–40 Gy** → complete necrosis without collateral damage.

- **Multi-institutional retrospective cohort** (localised tumours, median 5-year follow-up): SABR **safe**; local recurrence comparable to TA, but **CSS lower than expected** — below the near-100% 5-year CSS of active surveillance series — possibly reflecting a higher-risk population.
- **Limitations of that cohort**: non-RCC tumours included in survival analysis; heterogeneous single- and multi-fraction protocols without central, time-point-standardised review; highly selected comorbid population with substantial non-RCC mortality → recurrence likely underestimated.
- **Unresolved questions**: optimal dose fractionation (trials ongoing); **imaging endpoint** — SABR causes only subtle change, absence of growth is used, yet many untreated small masses barely grow; **post-SABR biopsies show viable but attenuated tumour** of unknown potential; need for larger prospective studies with long follow-up in populations comparable to surgery and TA.
- **Roles**: possibly combined with TA for larger tumours; **salvage after TA or PN** in comorbid patients; modern responsiveness contradicts the radioresistant reputation.
- **Advantages**: reduced morbidity; **no general anaesthesia**; **no interruption of anticoagulation**.
- **Status for small renal tumours**: **experimental** until dose fractionation, imaging criteria and prospective comparison with TA or surgery are established.

Irreversible Electroporation

- **Irreversible electroporation (IRE)**: novel **non-thermal** ablation; an electric field creates **nanoscale membrane pores**, reversible or lethally irreversible according to voltage.
- Delivered as electrical pulses via a single bipolar or multiple monopolar electrodes → reproducible cell death while **preserving extracellular matrix, tissue scaffolding, ductal structures and large vessels**.
- **Outcomes**: few studies in the past decade; **safe (no major complications)** but **oncologically suboptimal**, with frequent salvage for residual tumour.
- **Status**: **investigational**.

COMPARISON: INVESTIGATIONAL FOCAL MODALITIES

	HIFU	Histotripsy	SABR	IRE
Mechanism	Focused ultrasound; thermal + cavitation	Focused ultrasound; mechanical, non-thermal	High-dose stereotactic radiation	Electrical pulses; non-thermal pore formation
Invasiveness	Entirely extracorporeal	Extracorporeal	Non-invasive; no anaesthesia	Percutaneous electrodes
Key limitation	Incomplete ablation (respiration, acoustic interference)	No clinical data yet	No imaging endpoint; viable attenuated tumour; lower CSS	Suboptimal oncological control
Status	All investigational/experimental for small renal masses			

KEY POINTS: NEW ABLATION TECHNOLOGIES

- Several **promising investigational** focal modalities are emerging.
- **Stereotactic radiotherapy** may offer theoretical benefits, but substantial study is needed before any advantage over established TA can be claimed.

Concept — Each newer modality promises to escape a limitation of thermal ablation — needles, heat sink, collateral structural injury — but none yet has a validated imaging endpoint or outcome data matching the established trio, so all remain investigational.

CONCLUSION

- **TA is a guideline-endorsed alternative** for clinically localised small RCC; CA, RFA and MWA are standard, with **no prospective evidence that any one is superior**.
- **Local recurrence is higher than after surgery but usually salvaged by re-ablation**; meta-analyses show **no difference in MFS or CSS**.
- **Percutaneous ablation has less morbidity** than open or minimally invasive surgery with **renal function comparable to PN**.
- **Pretreatment biopsy** improves informed consent and shared decision making.
- Decision integrates **tumour characteristics, comorbidity, team experience and patient perspective**.